

ERA 5 calibration to Elexon power

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2026-07-09

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Overview

Updates

New

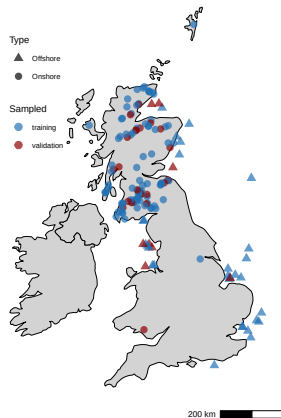
- OOS assessment
- Low wind events distribution

Model validation framework

Framework for OOS assessment

- Run models on a sample of days
 - 80/20 WF split for training and testing
 - Training with 3 days to predict the 4th
 - Randomly chosen 15 days from 2025, 5 per regime
- Evaluate performance at WF level
- Error under aggregation
 - GB level
 - Onshore / offshore

Sampled days for performance assessment

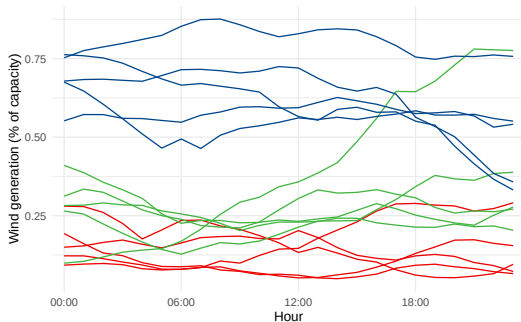


Sampled days in different regimes

- Objective is to assess performance by regime
- OOS days are different regimes of wind speed
- Previous 3 days can be anything

Sampled days for performance assessment

Daily generation time series for sampled days



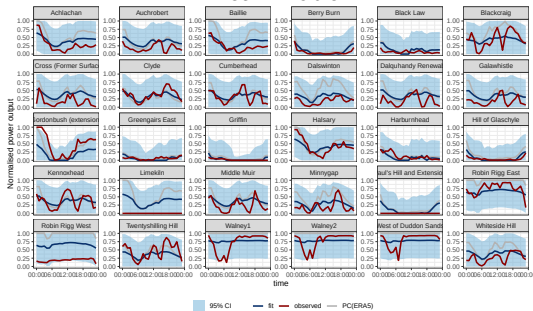
Model portfolio

- Wind farm level models:
 - Generic Power Curve (ERA5 + generic PC estimate)
 - Quantile Mapping
 - Linear model
 - AR1, AR2
 - Spatiotemporal model fine mesh and coarser mesh
- Linear model at Great Britain (GB) level

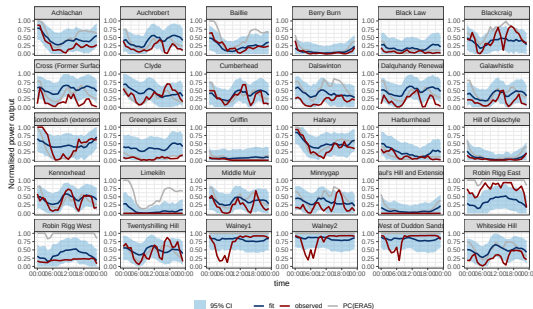
Out-of-sample time

Wind farm prediction bands

Linear model

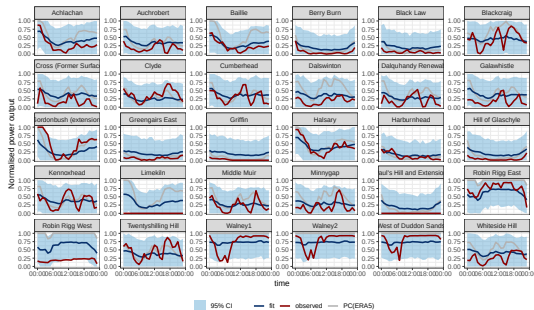


Spatiotemporal model coarse

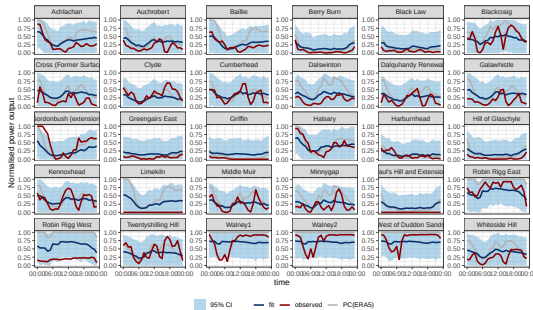


Wind farm prediction bands

1D SPDE model

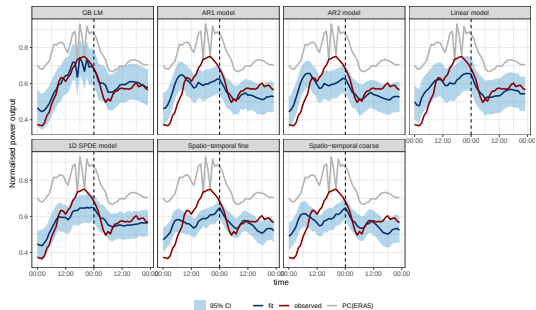


AR1 model



Aggregated prediction bands

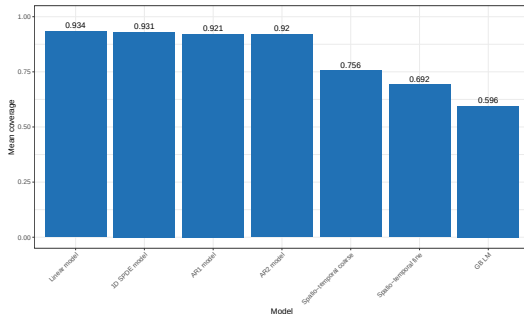
GB level



Day-ahead forecast coverage

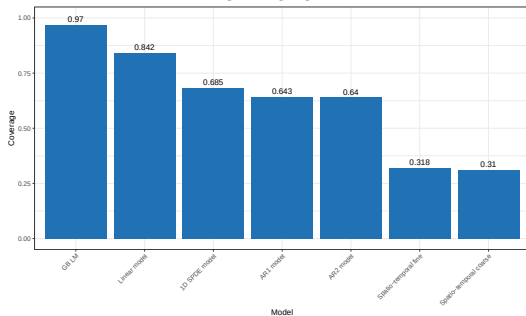
- Models at WF can achieve target coverage
- Aggregation reduces coverage
- High regime more susceptible to coverage loss
- Spatiotemporal model reduces coverage

Coverage for day-ahead predictions

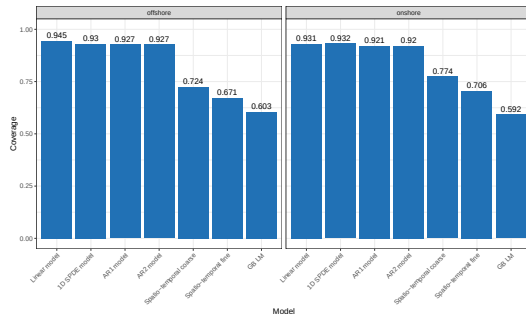


Coverage under aggregation

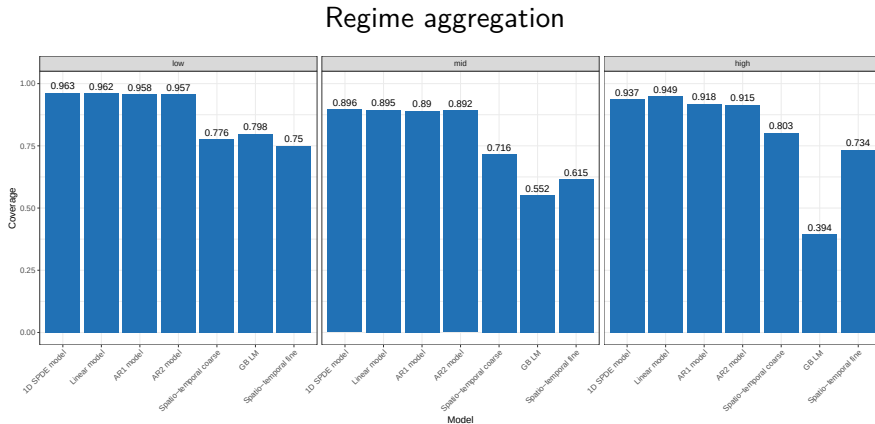
GB level



Onshore / offshore



Coverage under aggregation by regime



Out-of-sample locations